# Name: Muhammad Abu Bakar

# NUID: 24K-0826

# Section: BCS-243D

# Syllabus: Database Systems Lab

# Syllabus Code: CL2005

# Instructor: Sir Osama Tahir

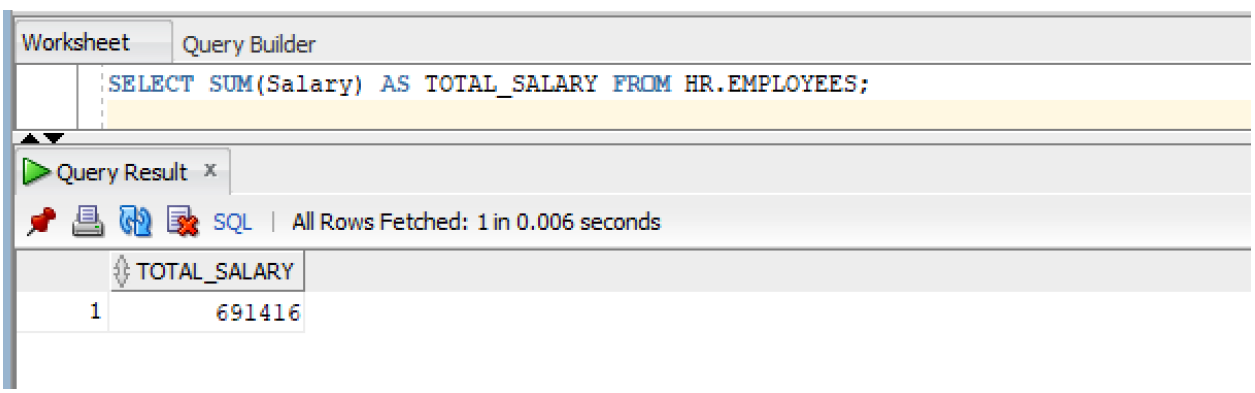
**Q1)**

**1)**

**SQL Query:**

SELECT SUM(Salary) AS TOTAL_SALARY FROM HR.EMPLOYEES;

**Snapshot of Query:**

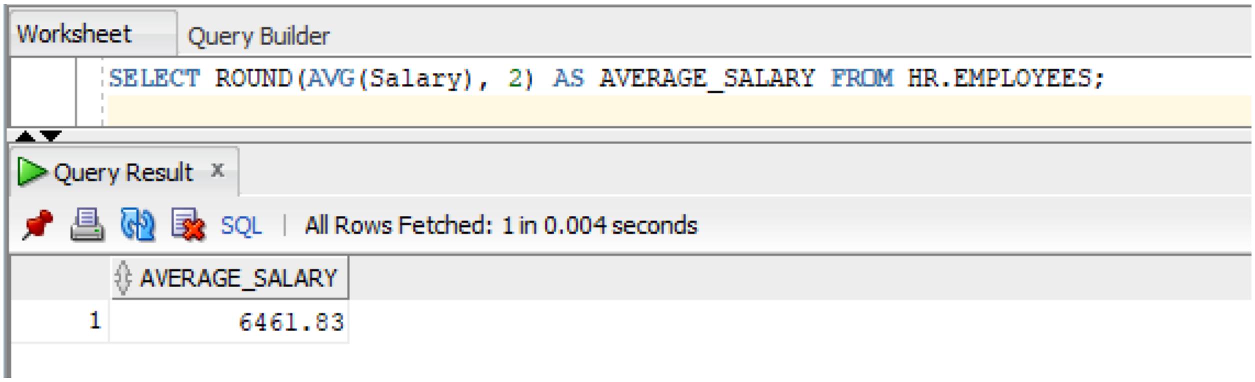


**2)**

**SQL Query:**

SELECT ROUND(AVG(Salary), 2) AS AVERAGE_SALARY FROM HR.EMPLOYEES;

**Snapshot of Query:**



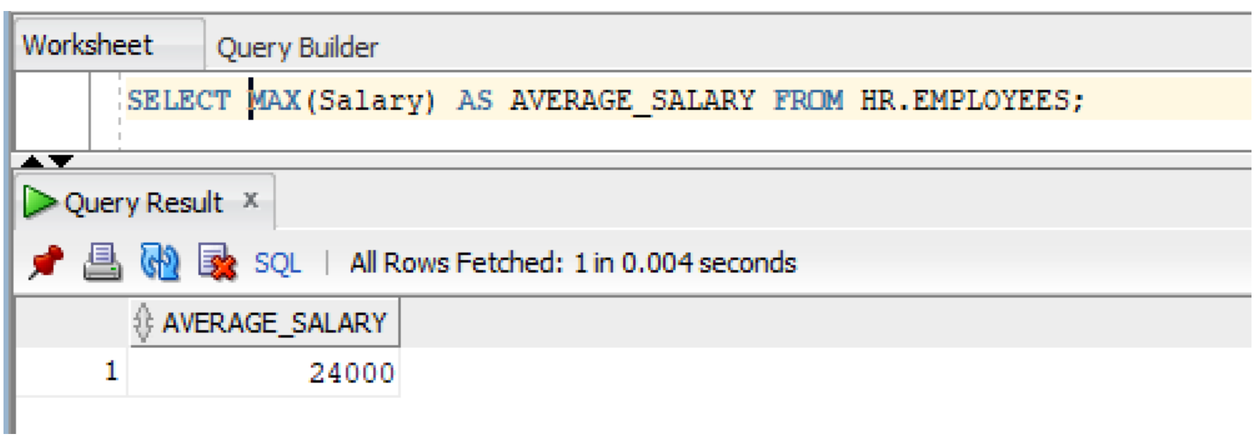

**3)**

**MAX**

**SQL Query:**

SELECT MAX(Salary) AS AVERAGE_SALARY FROM HR.EMPLOYEES;

**Snapshot of Query:**

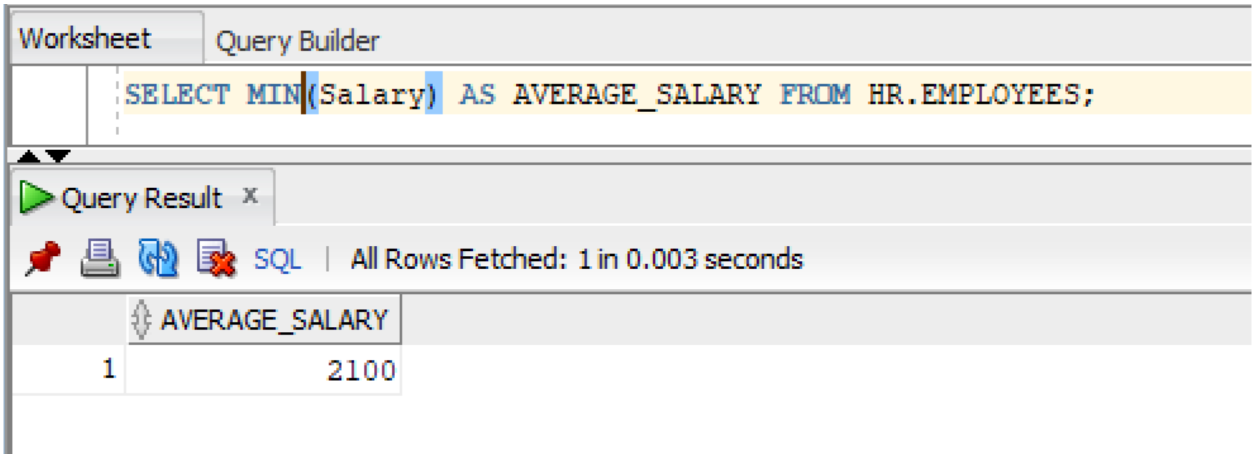


**MIN**

**SQL Query:**

SELECT MIN(Salary) AS AVERAGE_SALARY FROM HR.EMPLOYEES;

**Snapshot of Query:**

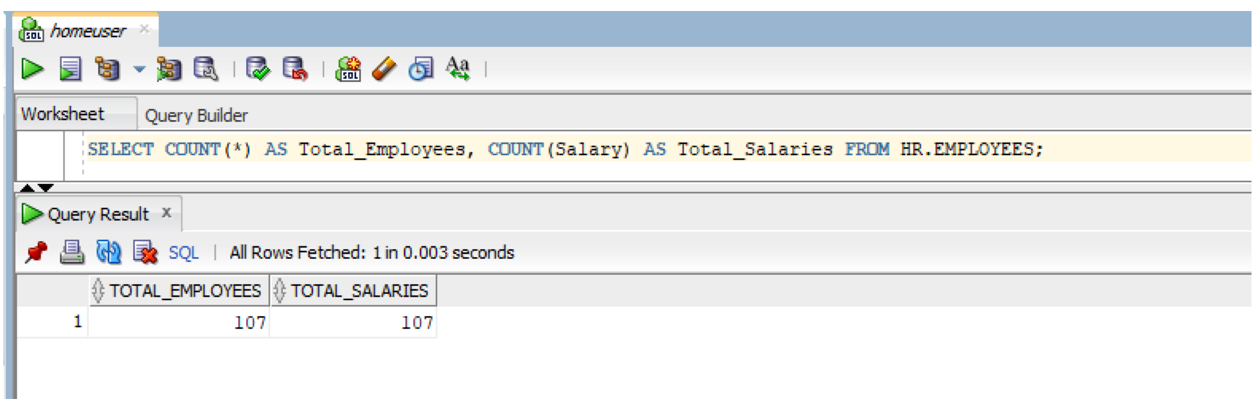



**4)**

**SQL Query:**

SELECT COUNT(*) AS Total_Employees, COUNT(Salary) AS Total_Salaries FROM HR.EMPLOYEES;

**Snapshot of Query:**

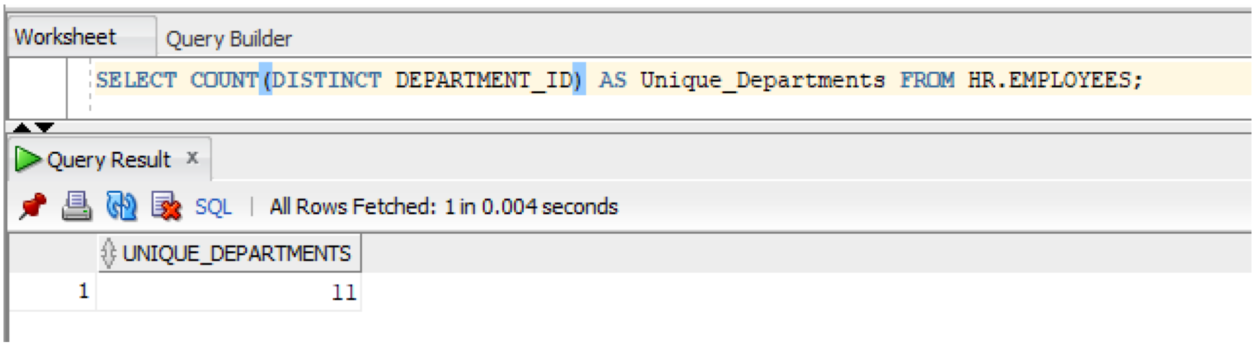


**5)**

## SQL Query:

SELECT COUNT(DISTINCT DEPARTMENT_ID) AS Unique_Departments FROM HR.EMPLOYEES;

## Snapshot of Query:



**Q2)**

**1)**

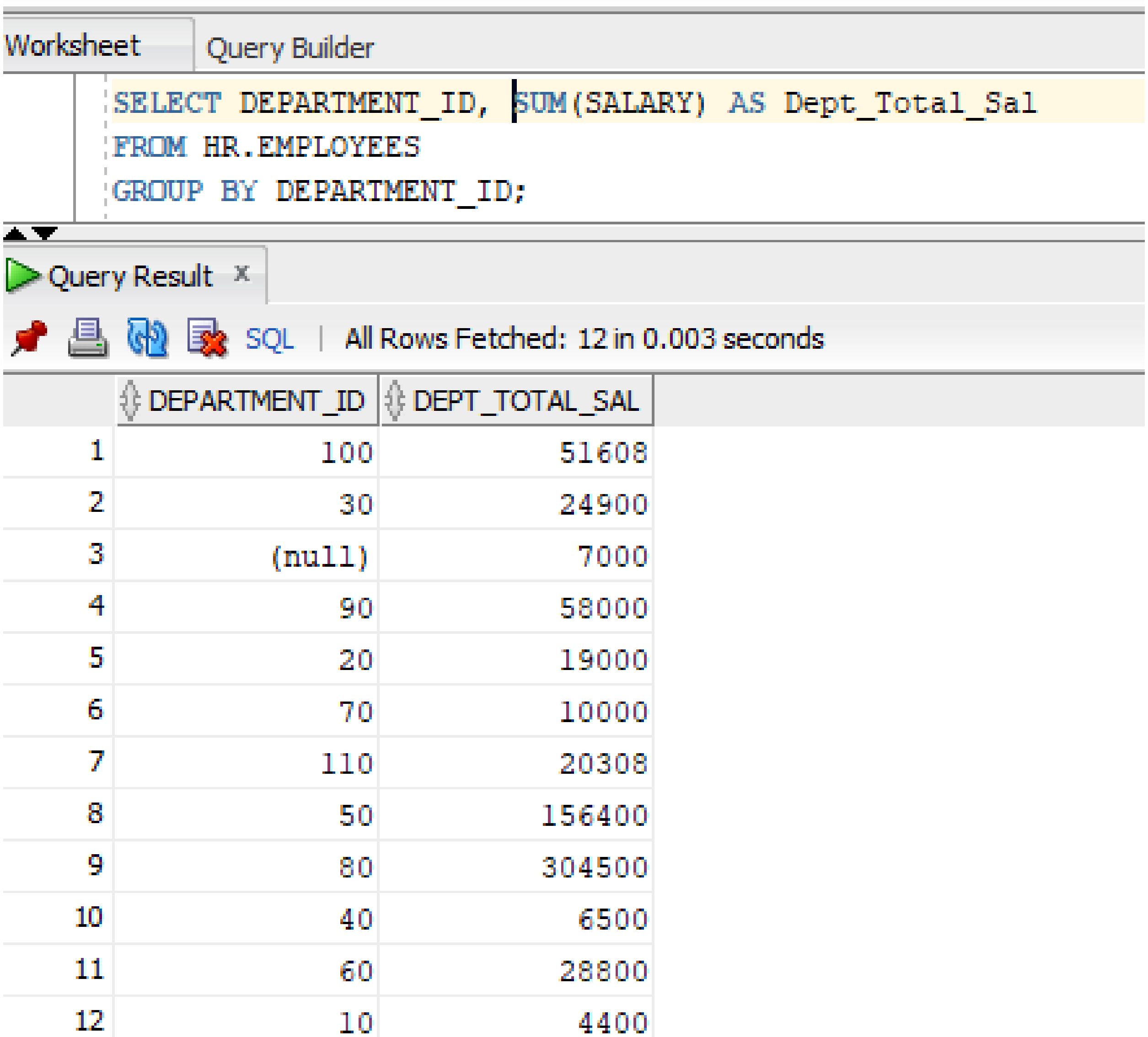
**SQL Query:**

SELECT DEPARTMENT_ID, SUM(SALARY) AS Dept_Total_Sal

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID;

**Snapshot of Query:**



Worksheet Query Builder

```
SELECT DEPARTMENT_ID, SUM(SALARY) AS Dept_Total_Sal
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID;
```

Query Result ×

SQL | All Rows Fetched: 12 in 0.003 seconds

| | DEPARTMENT_ID | DEPT_TOTAL_SAL |
|---|---|---|
| 1 | 100 | 51608 |
| 2 | 30 | 24900 |
| 3 | (null) | 7000 |
| 4 | 90 | 58000 |
| 5 | 20 | 19000 |
| 6 | 70 | 10000 |
| 7 | 110 | 20308 |
| 8 | 50 | 156400 |
| 9 | 80 | 304500 |
| 10 | 40 | 6500 |
| 11 | 60 | 28800 |
| 12 | 10 | 4400 |

2)

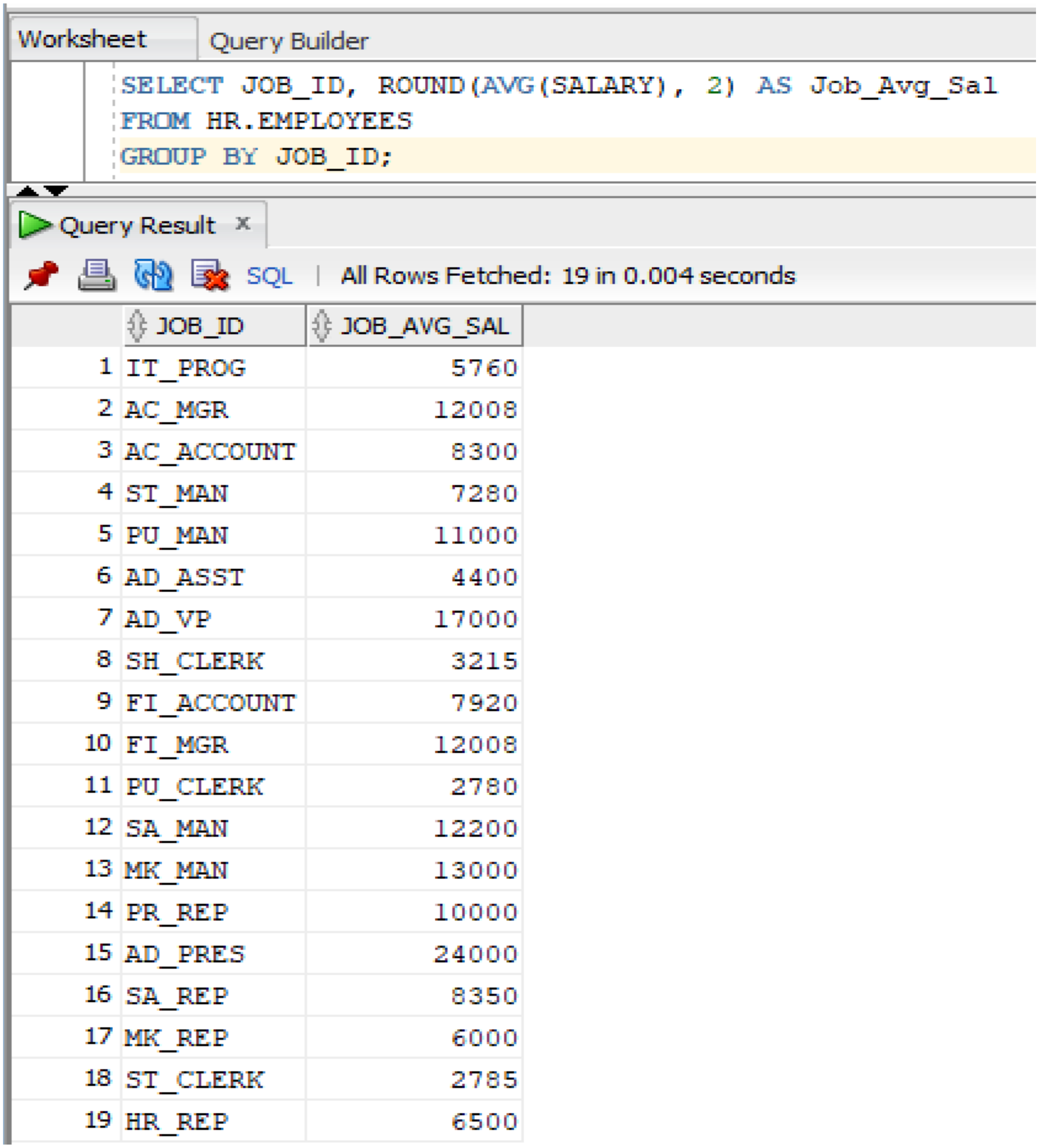
**SQL Query:**

SELECT JOB_ID, ROUND(AVG(SALARY), 2) AS Job_Avg_Sal

FROM HR.EMPLOYEES

GROUP BY JOB_ID;

**Snapshot of Query:**




| | JOB_ID | JOB_AVG_SAL |
|---|---|---|
| 1 | IT_PROG | 5760 |
| 2 | AC_MGR | 12008 |
| 3 | AC_ACCOUNT | 8300 |
| 4 | ST_MAN | 7280 |
| 5 | PU_MAN | 11000 |
| 6 | AD_ASST | 4400 |
| 7 | AD_VP | 17000 |
| 8 | SH_CLERK | 3215 |
| 9 | FI_ACCOUNT | 7920 |
| 10 | FI_MGR | 12008 |
| 11 | PU_CLERK | 2780 |
| 12 | SA_MAN | 12200 |
| 13 | MK_MAN | 13000 |
| 14 | PR_REP | 10000 |
| 15 | AD_PRES | 24000 |
| 16 | SA_REP | 8350 |
| 17 | MK_REP | 6000 |
| 18 | ST_CLERK | 2785 |
| 19 | HR_REP | 6500 |

**3)**

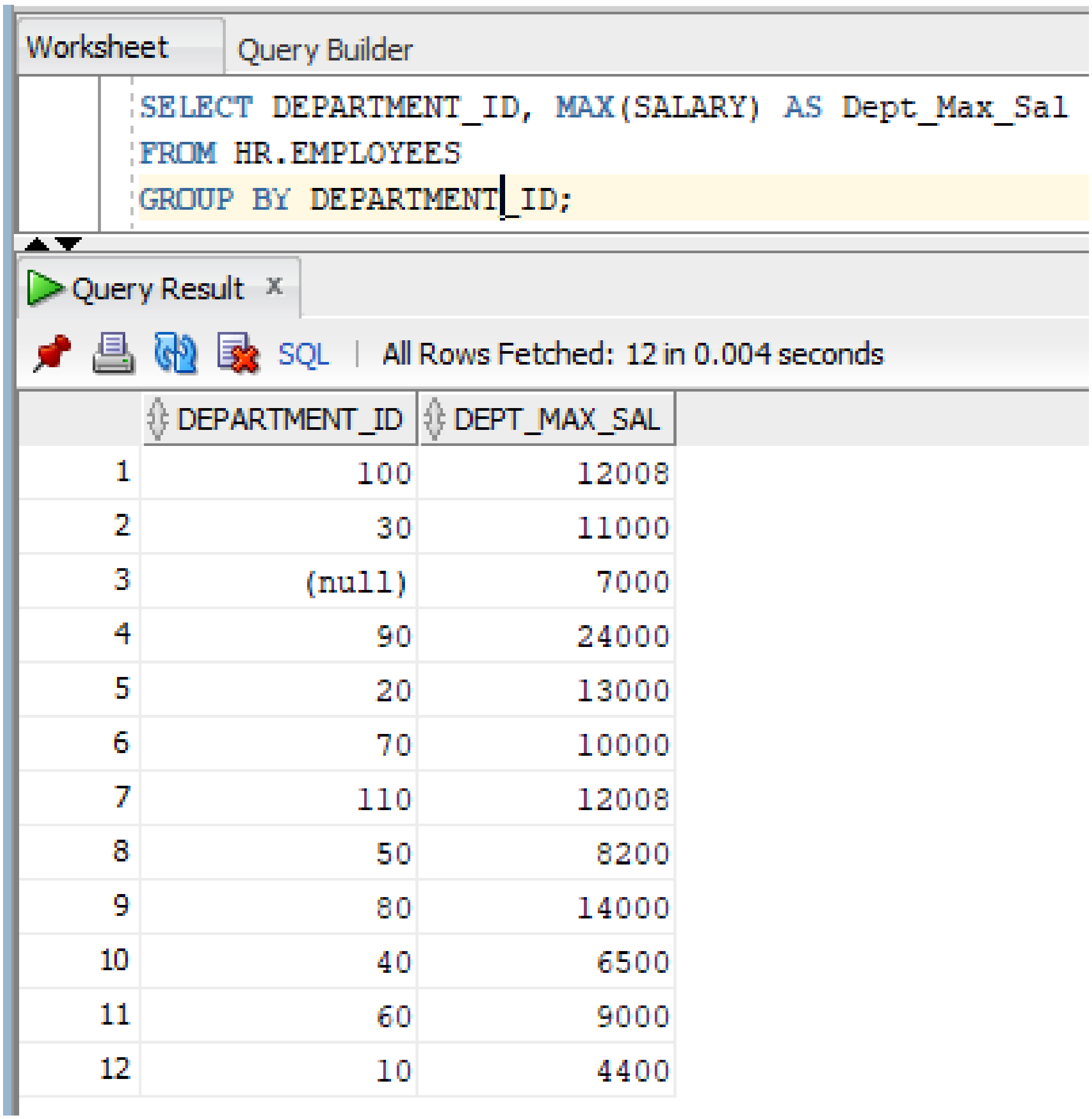
**SQL Query:**

SELECT DEPARTMENT_ID, MAX(SALARY) AS Dept_Max_Sal

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID;

**Snapshot of Query:**




| | DEPARTMENT_ID | DEPT_MAX_SAL |
|---|---|---|
| 1 | 100 | 12008 |
| 2 | 30 | 11000 |
| 3 | (null) | 7000 |
| 4 | 90 | 24000 |
| 5 | 20 | 13000 |
| 6 | 70 | 10000 |
| 7 | 110 | 12008 |
| 8 | 50 | 8200 |
| 9 | 80 | 14000 |
| 10 | 40 | 6500 |
| 11 | 60 | 9000 |
| 12 | 10 | 4400 |

**4)**

**SQL Query:**

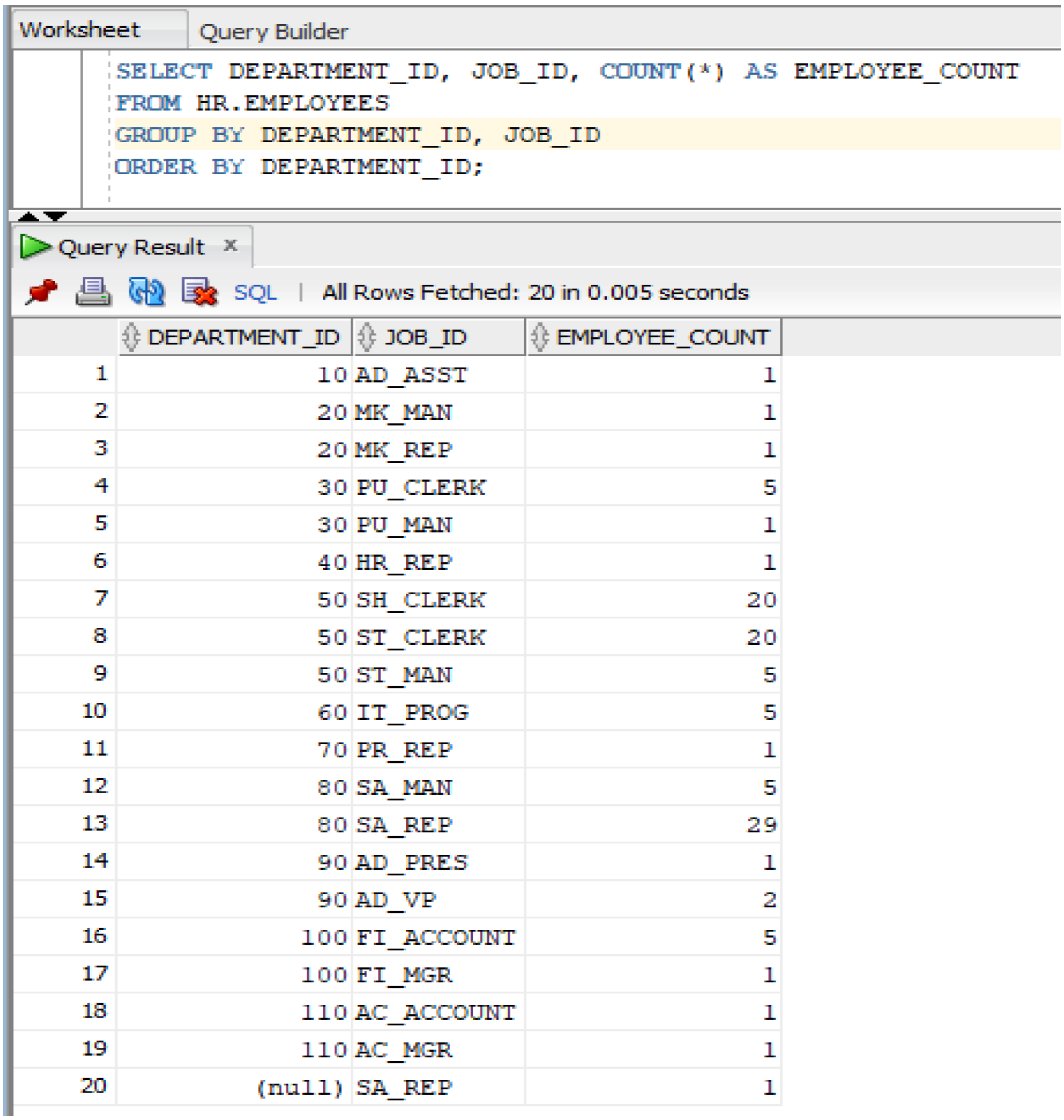
SELECT DEPARTMENT_ID, JOB_ID, COUNT(*) AS EMPLOYEE_COUNT

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID, JOB_ID

ORDER BY DEPARTMENT_ID;

**Snapshot of Query:**



Worksheet Query Builder

```
SELECT DEPARTMENT_ID, JOB_ID, COUNT(*) AS EMPLOYEE_COUNT
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID, JOB_ID
ORDER BY DEPARTMENT_ID;
```

Query Result ×

SQL | All Rows Fetched: 20 in 0.005 seconds

| | DEPARTMENT_ID | JOB_ID | EMPLOYEE_COUNT |
|---|---|---|---|
| 1 | 10 | AD_ASST | 1 |
| 2 | 20 | MK_MAN | 1 |
| 3 | 20 | MK_REP | 1 |
| 4 | 30 | PU_CLERK | 5 |
| 5 | 30 | PU_MAN | 1 |
| 6 | 40 | HR_REP | 1 |
| 7 | 50 | SH_CLERK | 20 |
| 8 | 50 | ST_CLERK | 20 |
| 9 | 50 | ST_MAN | 5 |
| 10 | 60 | IT_PROG | 5 |
| 11 | 70 | PR_REP | 1 |
| 12 | 80 | SA_MAN | 5 |
| 13 | 80 | SA_REP | 29 |
| 14 | 90 | AD_PRES | 1 |
| 15 | 90 | AD_VP | 2 |
| 16 | 100 | FI_ACCOUNT | 5 |
| 17 | 100 | FI_MGR | 1 |
| 18 | 110 | AC_ACCOUNT | 1 |
| 19 | 110 | AC_MGR | 1 |
| 20 | (null) | SA_REP | 1 |

**Q3)**

**1)**

**SQL Query:**

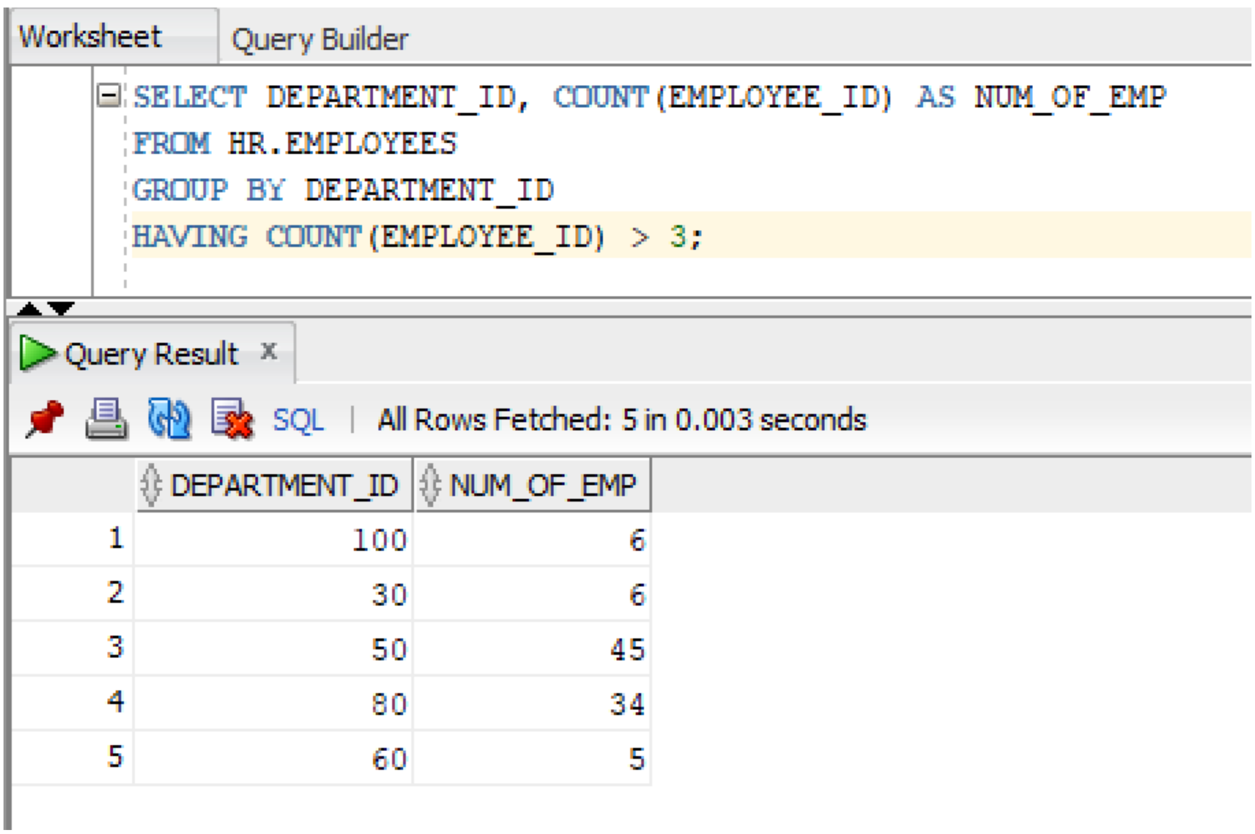
SELECT DEPARTMENT_ID, COUNT(EMPLOYEE_ID) AS NUM_OF_EMP

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID

HAVING COUNT(EMPLOYEE_ID) > 3;

**Snapshot of Query:**



2)

**SQL Query:**

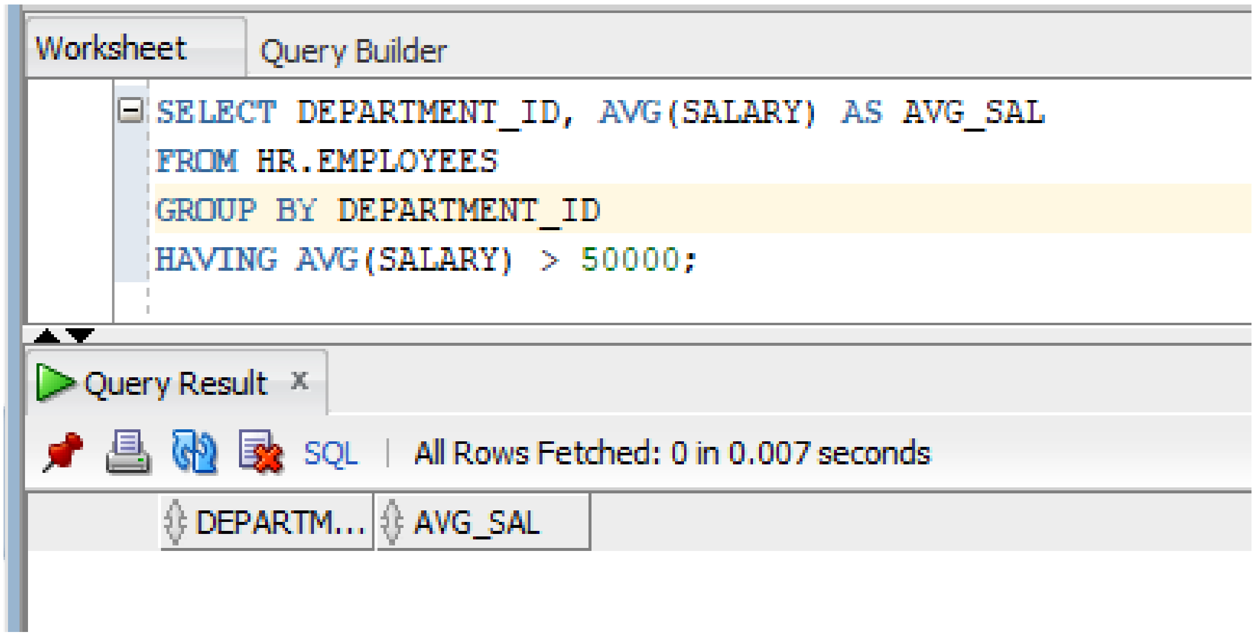
SELECT DEPARTMENT_ID, AVG(SALARY) AS AVG_SAL

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID

HAVING AVG(SALARY) > 50000;

**Snapshot of Query:**



**3)**

**SQL Query:**

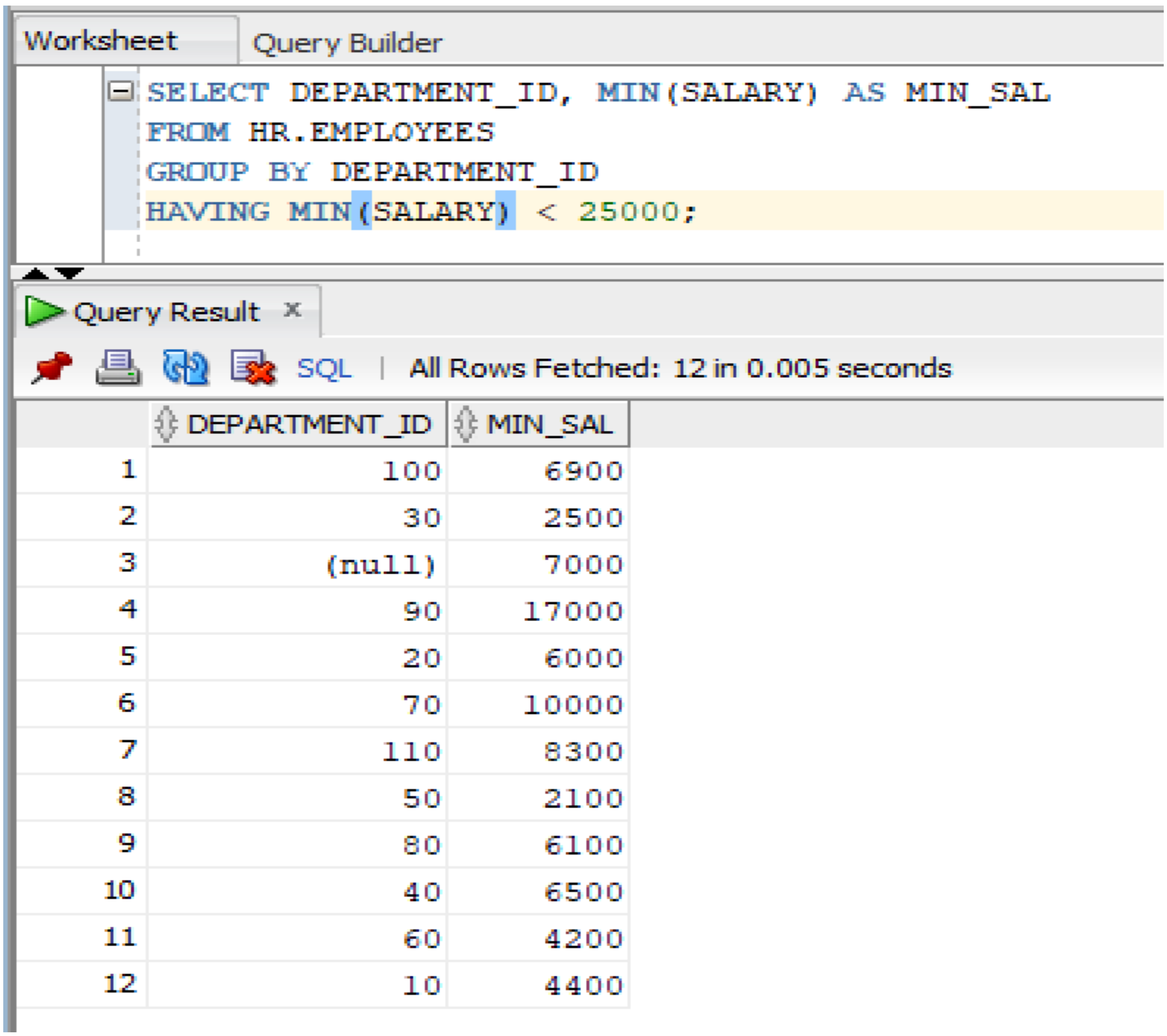
SELECT DEPARTMENT_ID, MIN(SALARY) AS MIN_SAL

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID

HAVING MIN(SALARY) < 25000;

**Snapshot of Query:**




| | DEPARTMENT_ID | MIN_SAL |
|---|---|---|
| 1 | 100 | 6900 |
| 2 | 30 | 2500 |
| 3 | (null) | 7000 |
| 4 | 90 | 17000 |
| 5 | 20 | 6000 |
| 6 | 70 | 10000 |
| 7 | 110 | 8300 |
| 8 | 50 | 2100 |
| 9 | 80 | 6100 |
| 10 | 40 | 6500 |
| 11 | 60 | 4200 |
| 12 | 10 | 4400 |

**Q4)**

**1)**

**SQL Query:**

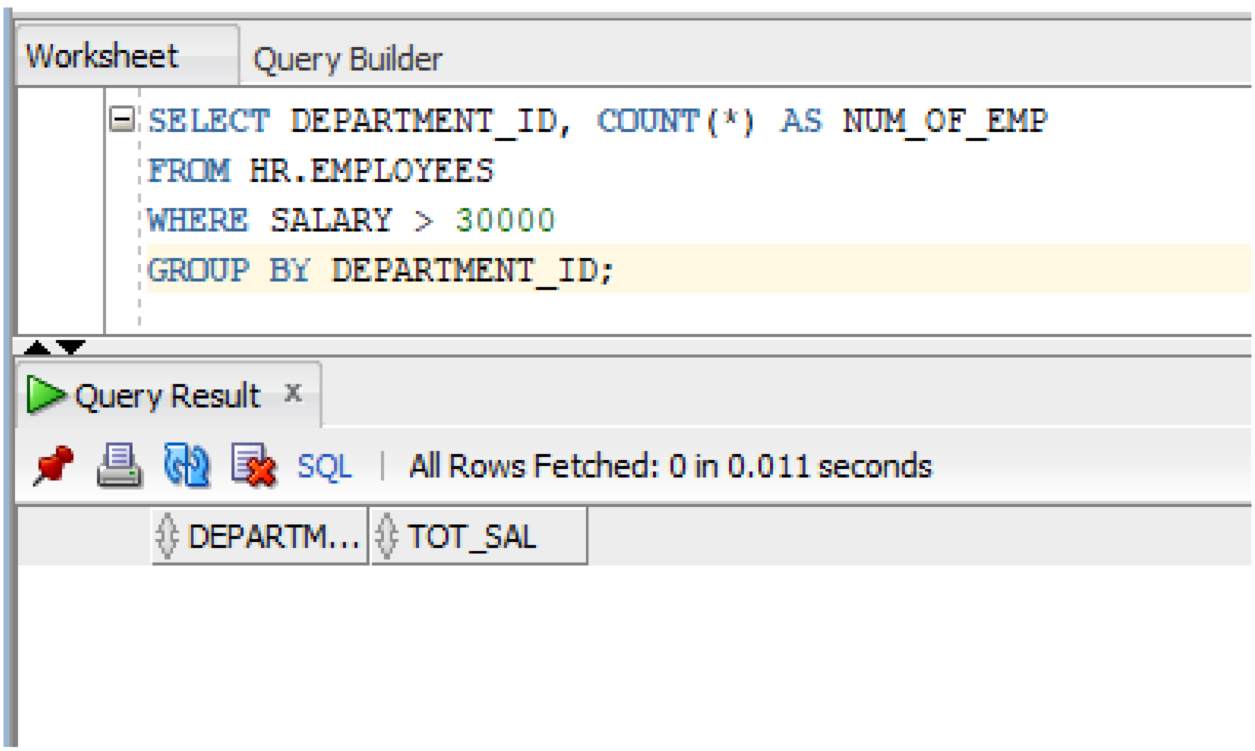
SELECT DEPARTMENT_ID, COUNT(*) AS NUM_OF_EMP

FROM HR.EMPLOYEES

WHERE SALARY > 30000

GROUP BY DEPARTMENT_ID;

**Snapshot of Query:**



**2)**

**SQL Query:**

SELECT DEPARTMENT_ID, SUM(SALARY) AS TOT_SAL

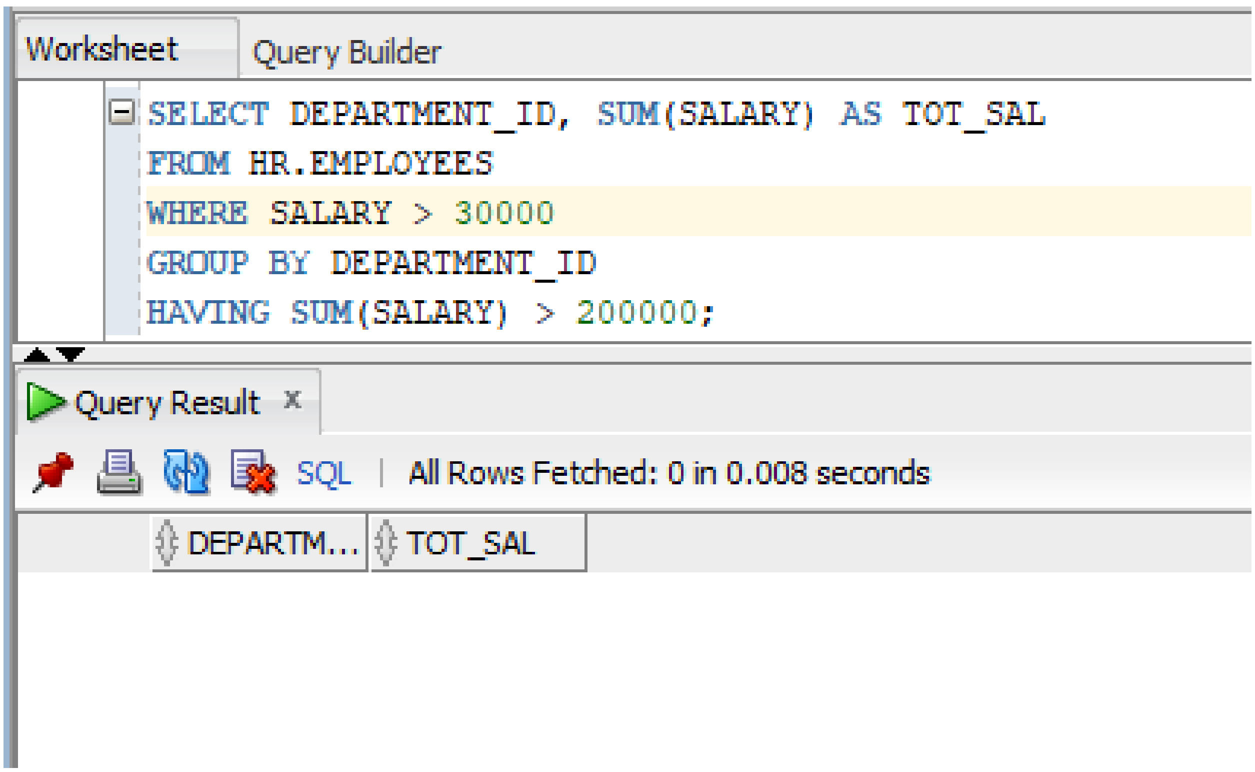
FROM HR.EMPLOYEES

WHERE SALARY > 30000

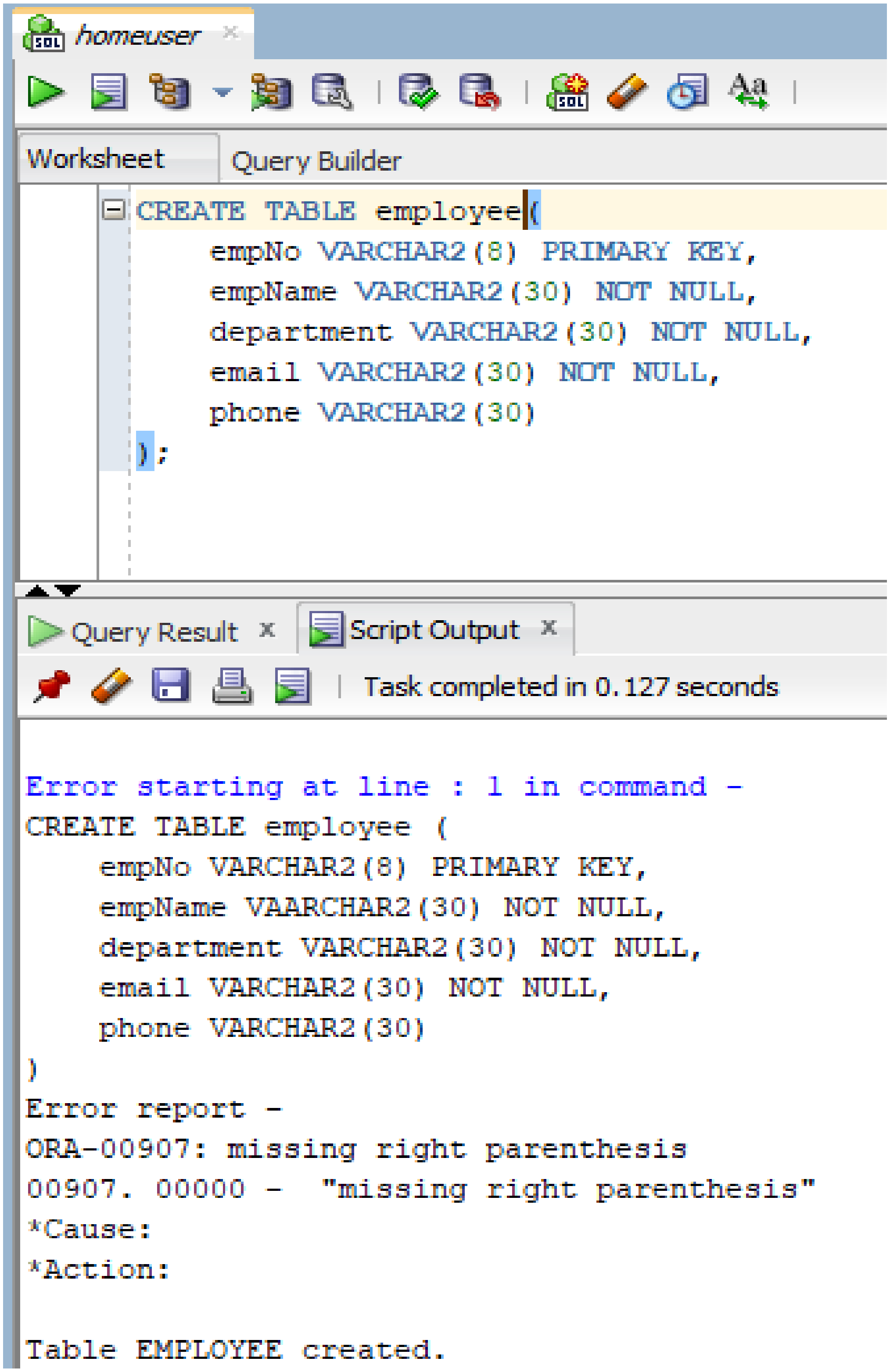
GROUP BY DEPARTMENT_ID

HAVING SUM(SALARY) > 200000;

**Snapshot of Query:**



**PART B – DML OPERATIONS**



**Q5)**

## SQL Query:

INSERT INTO employee (empNo, empName, department, email, phone)

VALUES ('E001', 'Muhammad Abu Bakar', 'IT', 'bakar@gmail.com', '0300-1234567');

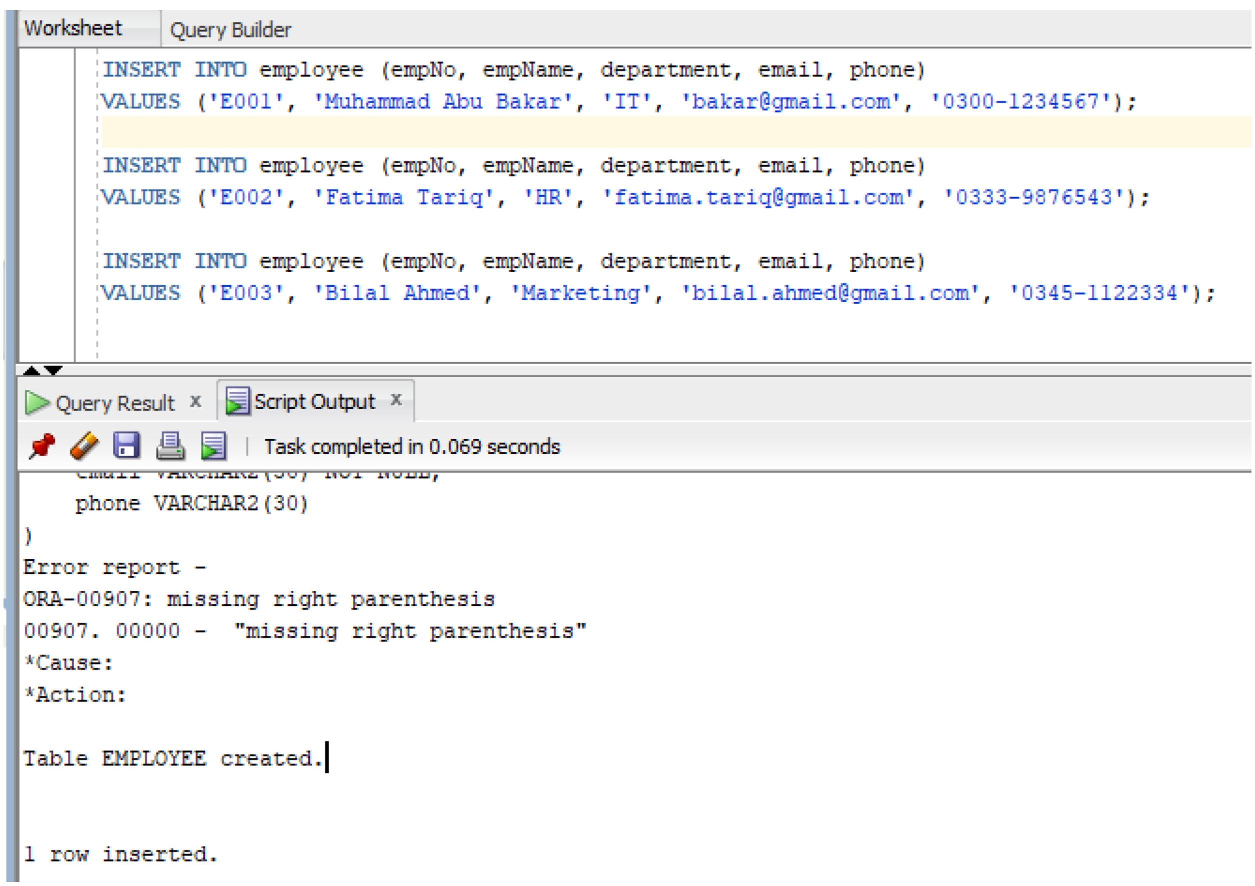
INSERT INTO employee (empNo, empName, department, email, phone)

VALUES ('E002', 'Fatima Tariq', 'HR', 'fatima.tariq@gmail.com', '0333-9876543');

INSERT INTO employee (empNo, empName, department, email, phone)

VALUES ('E003', 'Bilal Ahmed', 'Marketing', 'bilal.ahmed@gmail.com', '0345-1122334');

## Snapshot of Query:



**Q6)**

**1)**

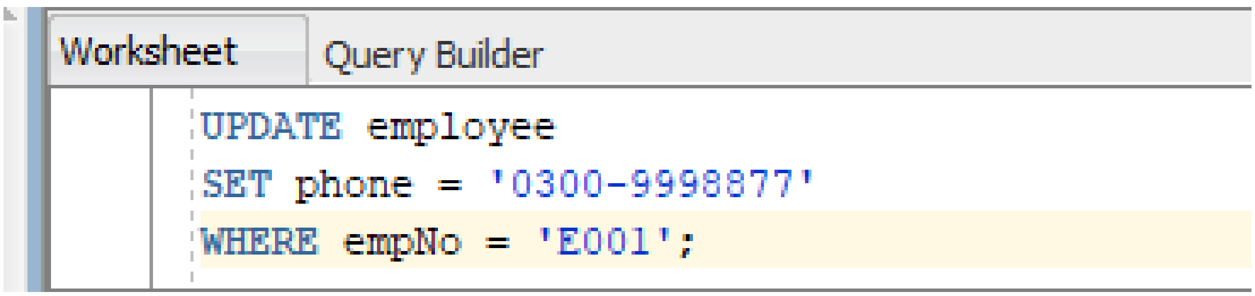
**SQL Query:**

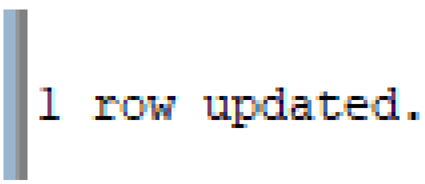
UPDATE employee

SET phone = '0300-9998877'

WHERE empNo = 'E001';

**Snapshot of Query:**



**2)**

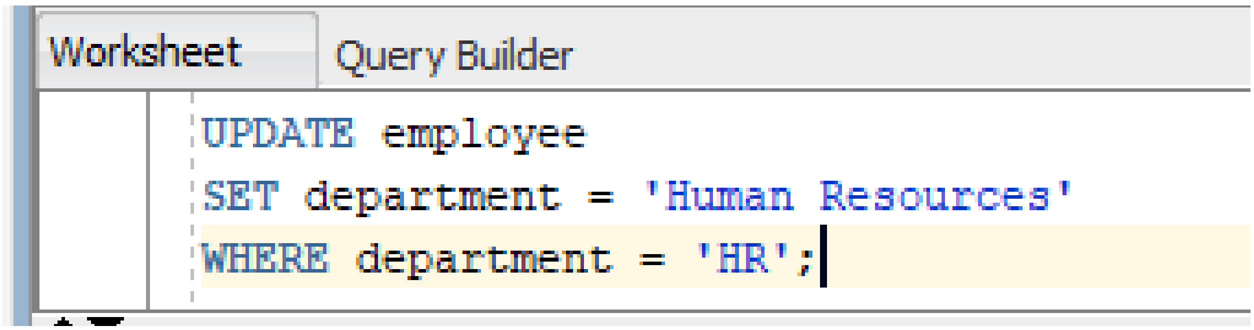
**SQL Query:**

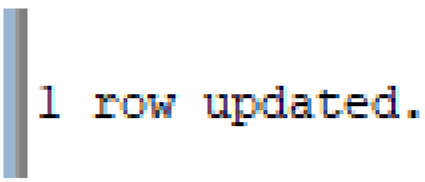
UPDATE employee

SET department = 'Human Resources'

WHERE department = 'HR';

**Snapshot of Query:**



3)

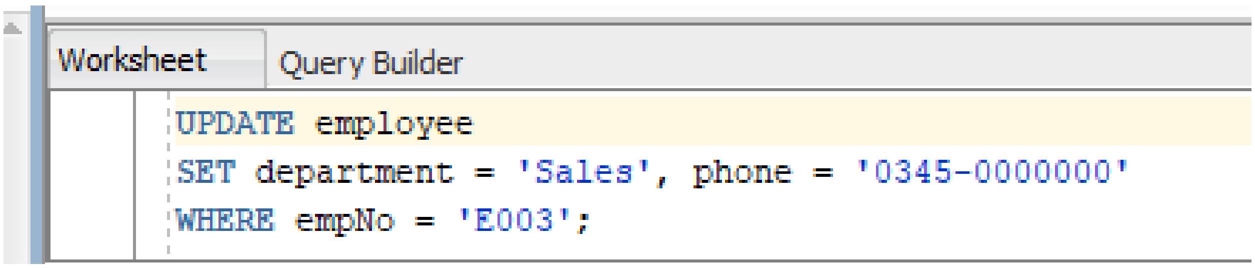
**SQL Query:**

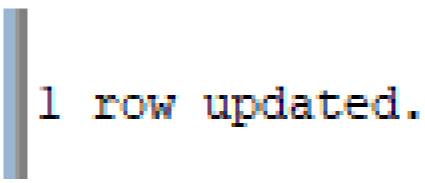
UPDATE employee

SET department = 'Sales', phone = '0345-0000000'

WHERE empNo = 'E003';

**Snapshot of Query:**



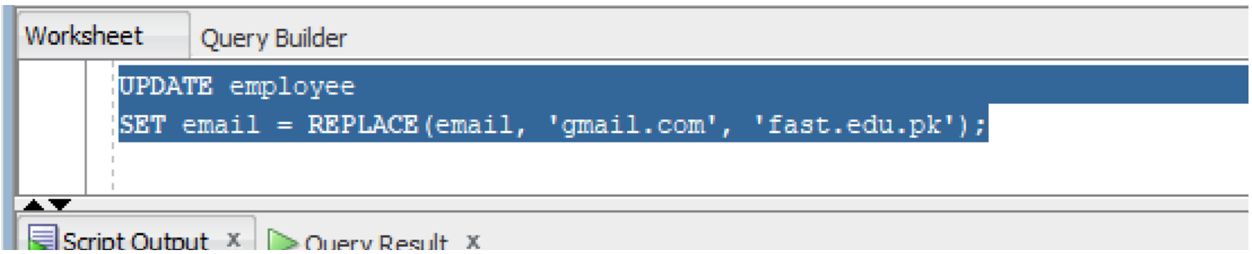
4)

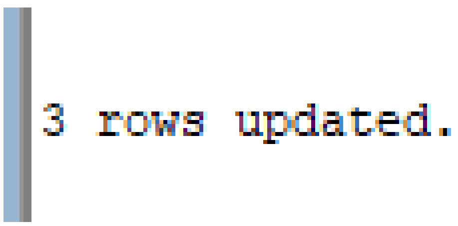
**SQL Query:**

UPDATE employee

SET email = REPLACE(email, 'gmail.com', 'fast.edu.pk');

**Snapshot of Query:**



**Q7)**

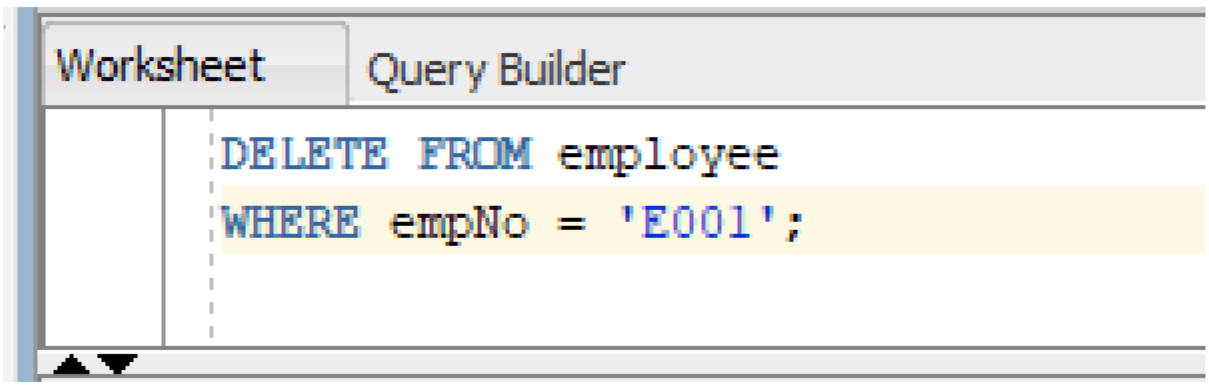
**1)**

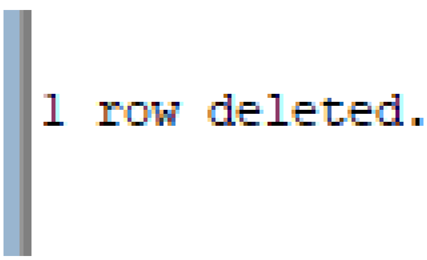
**SQL Query:**

DELETE FROM employee

WHERE empNo = 'E001';

**Snapshot of Query:**



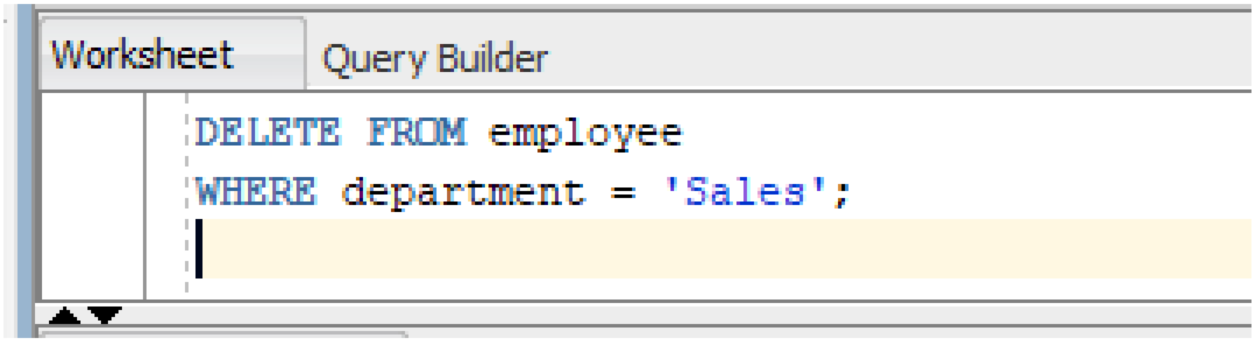
**2)**

**SQL Query:**

DELETE FROM employee

WHERE department = 'Sales';

**Snapshot of Query:**




**3)**

**SQL Query:**

DELETE FROM employee;

**Snapshot of Query:**

Worksheet Query Builder

```sql
DELETE FROM employee;
```

Script Output × Query Result ×

Task completed in 0.05 seconds

```
Error starting at line : 7 in command -
INSERT INTO employee (empNo, empName, department, email, phone)
VALUES ('E003', 'Bilal Ahmed', 'Marketing', 'bilal.ahmed@gmail.com', '0345-1122334')
Error report -
ORA-00001: unique constraint (SYS.SYS_C007215) violated


1 row inserted.


1 row inserted.


1 row updated.


1 row updated.


1 row updated.


3 rows updated.


1 row deleted.


2 rows deleted.
```

**4)**

**SQL Query:**

TRUNCATE TABLE tasks;

**Snapshot of Query:**